

Deep Learning-Based Prediction of Trabecular Bone Mineral Density From Lumbar CT: A Superior Alternative to DEXA in Hyperostosis

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Abstract

Study Design: Retrospective Cohort Study.

Objectives: This study explores using deep learning to predict lumbar spine bone mineral density (BMD) from CT images, aiming to improve osteoporosis and osteopenia detection and facilitate early intervention. While dual-energy X-ray absorptiometry (DEXA) is the gold standard, it often overestimates BMD in skeletal hyperostosis. We sought to develop a CT-based model capable of accurately predicting trabecular BMD and validating its fracture-risk prediction performance.

Methods: This retrospective study analyzed 1840 lumbar vertebrae (L1–L4) from 460 patients with available DEXA results. Patients were randomly split 7:3 (patient-level) into training and test cohorts. Trabecular regions were semi-automatically segmented, and based on DEXA, we developed a regression model for predicting BMD values. Additionally, a three-class deep learning model classified normal bone mass, osteopenia, and osteoporosis. Next, we established a BMD–fracture risk prediction model in patients without lumbar osteophytosis or ligamentous calcification and externally validated its performance. In patients with lumbar osteophytosis or ligamentous calcification, we compared fracture-risk prediction between DEXA and model-predicted BMD.

Results: Model-predicted BMD correlated strongly with DEXA ($r = 0.934$). The three-class model achieved AUCs of 0.90 (normal), 0.79 (osteopenia), and 0.92 (osteoporosis). External validation confirmed that model-predicted BMD (AUC = 0.901) outperformed DEXA (AUC = 0.464) in fracture-risk prediction among hyperostotic patients.

Conclusions: Deep learning enabled accurate BMD estimation from lumbar CT images, supporting osteoporosis diagnosis and fracture-risk stratification. Notably, in patients with lumbar osteophytosis or ligamentous calcification, CT-based predicted BMD provided superior risk discrimination compared with DEXA, indicating closer alignment with actual bone status.

Keywords

bone density, deep learning, lumbar vertebrae, tomography, X-Ray computed, bone diseases, metabolic, osteoporosis

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Background

Osteoporosis is a systemic skeletal disorder marked by reduced bone mass and a deterioration of the microarchitecture of bone tissue¹; it leads to increased bone fragility and a high risk of fractures. Osteoporotic fractures can lead to significant pain, disability, and even death, creating a substantial societal burden.^{2,3} According to the Seventh National Population Census, 264 million people in China are aged 60 or older, accounting for 18.7% of the total population.⁴ Osteoporosis is becoming a major public health issue in China due to its rapidly increasing prevalence among the aging population.

DEXA (dual-energy X-ray absorptiometry) is commonly used to measure BMD (bone mineral density) in the spine and hip joints.⁵ DEXA calculates areal BMD by projecting the 3D bone structure onto a 2D plane, which includes both cortical and trabecular bones. However, it cannot exclude the impact of cortical bone, osteophytes, and sclerosis,⁶ potentially resulting in overestimated BMD and T-scores.^{7,8} A T-score represents the number of standard deviations by which a patient's BMD differs from the mean BMD of a young healthy reference population. According to the World Health Organization diagnostic criteria based on DEXA T-scores, osteoporosis is defined as a T-score ≤ -2.5 , osteopenia as $-2.5 < \text{T-score} < -1.0$, and normal bone density as a T-score ≥ -1.0 .⁹ Osteopenia represents low bone mass that is below normal but does not meet the diagnostic threshold for osteoporosis, and it is clinically important because patients with osteopenia may have an increased risk of fragility fractures.⁹ QCT (quantitative computed tomography) is recognized as a 3D method for evaluating bone density and demonstrates a greater ability to detect osteoporosis than DEXA.^{10,11} However, QCT requires calibration and standardized software, which complicates its use. Moreover, it delivers a significantly higher radiation dose than DEXA, restricting its suitability as a screening method.

The application of CT scans is becoming increasingly widespread, particularly in the abdominal and lumbar spine regions. Researchers have developed models to screen for osteoporosis and osteopenia using CT scans.^{12,13} These classification models determine osteoporosis and osteopenia. However, they have limited precision for continuous BMD value prediction or function as simplified binary classification models.¹³ Predicting bone density in one region using measurements from another is limited by the varying rates of bone loss across different anatomical sites with age, which restricts the model's clinical applicability in orthopedics.

Deep learning is being increasingly applied to radiological diagnosis,¹⁴ particularly in osteoporosis.^{12,15} However, previous studies that selected the entire vertebral body as the ROI (region of interest) faced challenges in avoiding the influence of cortical bone on actual bone density measurements. The density of vertebral trabecular bone is central to clinical decision-making and patient outcomes.^{16,17} The vertebral trabecular compartment, rather than its cortical shell, is the

primary determinant of both surgical planning and fracture risk.

This study developed a deep learning model to predict lumbar spine bone density from CT scans. Additionally, a three-class classification model was developed to support clinically meaningful screening and stratification of osteoporosis and osteopenia, thereby facilitating timely confirmatory assessment, referral for osteoporosis evaluation, and consideration of guideline-based management when appropriate. The models were trained using DEXA results from patients meeting strict inclusion and exclusion criteria, with the ROI carefully delineated to represent the trabecular bone, ensuring more accurate bone density assessment.

Additionally, we constructed a fracture-risk prediction model using non-hyperostotic fracture patients as an external validation cohort to determine whether DEXA-measured or model-predicted BMD more accurately predicts thoracolumbar fracture risk, thereby establishing the model's performance specifically within the hyperostotic population.

Methods

This retrospective study received approval from the Ethics Committee and Institutional Review Board of our hospital (2024-SR-132). Based on its retrospective design, informed consent was not required. To ensure patient privacy, the data were anonymized prior to use.

Patients

Images and data were collected from our hospital between January 2022 and September 2023. The criteria for inclusion were as follows: (1) patients aged from 18 to 75 years; (2) underwent DEXA. The criteria for exclusion were as follows: (1) patients who did not undergo a lumbar spine CT scan within 2 weeks of BMD measurement; (2) a history of surgery involving L1 to L4; (3) severe scoliosis or substantial lumbar spine deformity¹⁸; (4) tumors or tuberculosis lesions in the lumbar spine; (5) fractures of L1 to L4; (6) L1–L4 lumbar vertebral osteophytosis, calcification of the anterior longitudinal ligament, calcification of the posterior longitudinal ligament, and ossification or calcification of the ligamentum flavum. Lumbar hyperostosis and ligamentous calcification were evaluated on CT using pre-specified imaging thresholds. A positive finding was defined as the presence of either of the following at any level from L1 to L4: (1) lumbar spondylosis graded ≥ 2 according to the Kellgren–Lawrence (K–L) system on bone-window sagittal reconstructed CT images,¹⁹ or (2) definite ligamentous calcification involving the anterior longitudinal ligament, posterior longitudinal ligament, or ligamentum flavum. Two experienced readers independently performed the assessments while blinded to DEXA results and fracture grouping. Disagreements were resolved by consensus, with adjudication by a senior reader when necessary.

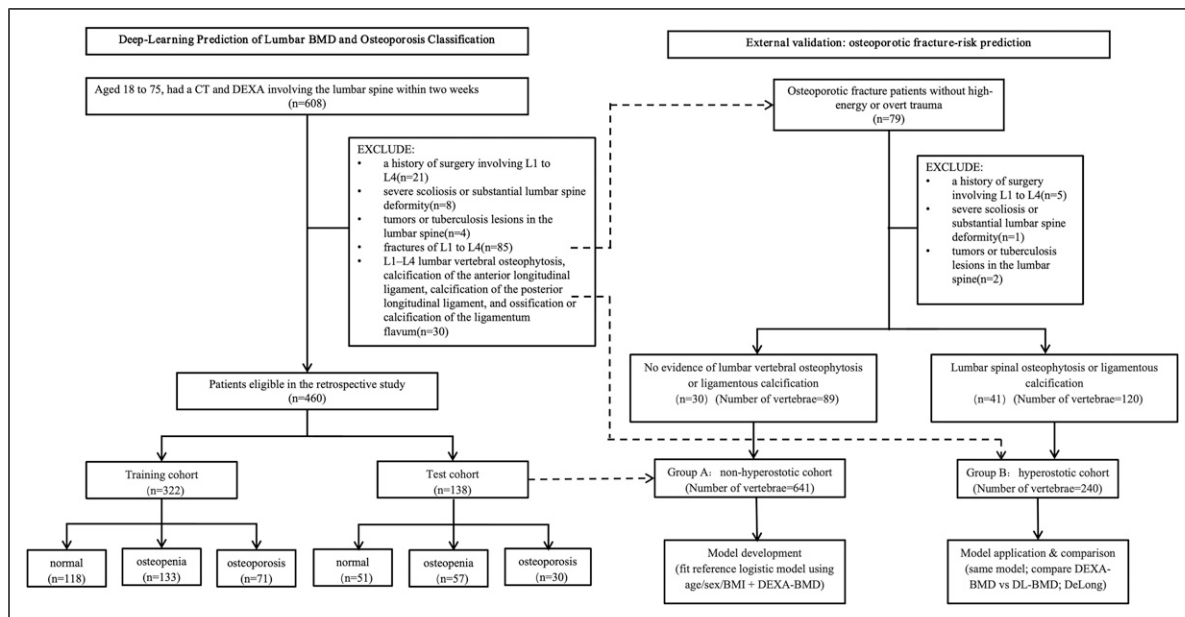


Figure 1. Flowchart for selecting the study population. Solid arrows depict the cohort stratification and grouping workflow used for model development and the external-validation analysis, whereas dashed arrows indicate the provenance of specific subgroups within the external-validation cohort

This retrospective study included 1840 lumbar vertebrae (L1–L4). These vertebrae were derived from 460 patients who met the inclusion and exclusion criteria. We performed a patient-level multivariable stratified random split in a 7:3 ratio to create the training and test cohorts. Stratification used prespecified factors including the WHO BMD diagnostic categories defined by DEXA T-scores (normal, osteopenia, osteoporosis) and key baseline characteristics (sex, age, height, weight, and BMI). All vertebrae from the same patient were assigned to the same cohort, thereby preventing data leakage. This yielded 1288 vertebrae in the training cohort and 552 vertebrae in the test cohort. Figure 1 illustrates the process of patient inclusion and exclusion in this study.

External Validation Cohort

Patients admitted to our spinal surgery unit for osteoporotic fractures between January 2022 and September 2023, in the absence of high-energy or overt trauma, were screened. The external-validation fracture cohort applied the same general eligibility criteria as the BMD-prediction cohort (lumbar CT within 2 weeks of DEXA, no prior L1–L4 surgery, no severe scoliosis/substantial lumbar deformity, and no lumbar tumor or tuberculosis), and patients with high-energy or overt trauma-related fractures were excluded. Unlike the BMD-prediction cohort, patients with lumbar osteophytosis or ligamentous calcification were retained for the fracture-risk analysis. After excluding fractured vertebrae to avoid fracture-related alterations in trabecular structure and ROI appearance that could bias BMD estimation, the retrospective

cohort comprised 71 eligible patients contributing 209 non-fractured L1–L4 vertebrae.

For the fracture-risk analysis, we compared two BMD inputs: DEXA-measured BMD and deep learning–predicted trabecular BMD. Participants were stratified according to the presence of lumbar hyperostosis (osteophytosis and/or ligamentous calcification) into Group A (non-hyperostotic) and Group B (hyperostotic).

In Group A, non-hyperostotic fracture patients ($n = 30$) were combined with non-hyperostotic controls from the validation subset of the BMD-prediction cohort (held out from model training) ($n = 138$) to develop the fracture-risk reference model using DEXA-BMD.

In Group B, hyperostotic fracture patients ($n = 41$) and hyperostotic controls without vertebral fracture ($n = 30$) screened under the same eligibility criteria were used to compare fracture-risk discrimination between DEXA-BMD and deep learning–predicted BMD by applying both modalities to the same reference model. Figure 1 summarizes cohort assembly and group definitions.

Image Acquisition

All CT images and DEXA results were obtained from our department’s Picture Archiving and Communication System. Each patient underwent a CT scan that included the lumbar spine, using eight different brands and models of 64-slice spiral CT scanners (Philips iCT256, Philips Incisive CT, Philips IQon-Spectral CT, GE MEDICAL SYSTEMS Optima CT660, SIMENS SOMA TOM Definition AS+, SIMENS

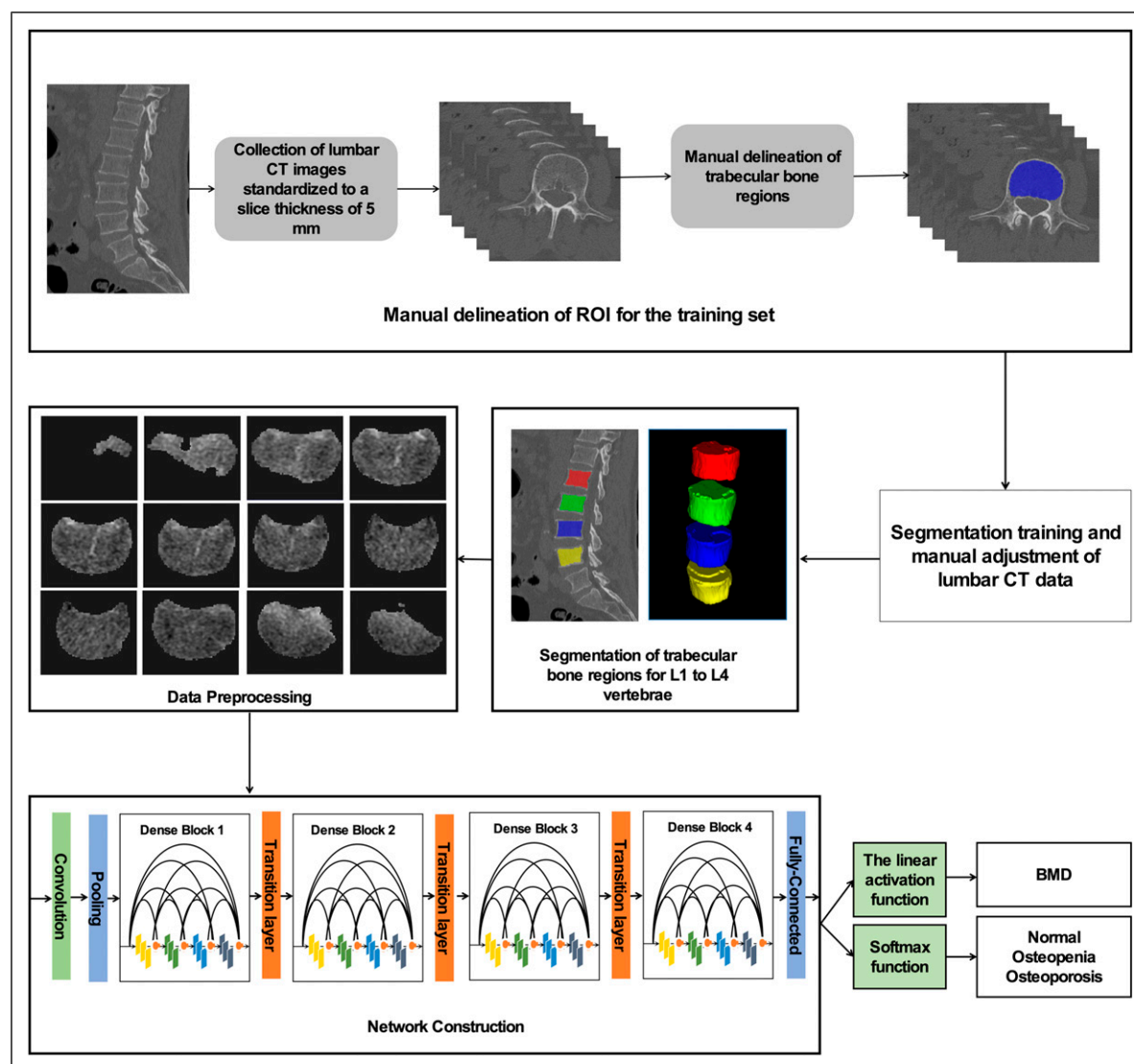


Figure 2. Semi-automatic trabecular ROI segmentation and DenseNet-121-based deep-learning workflow. Two independent models were trained separately: one for continuous trabecular BMD regression and the other for bone-status classification (normal/osteopenia/osteoporosis)

Emotion 16, SIEMENS Healthineers SOMATOM go.UP and UIH uCT 960+). The scanning parameters were set as follows: tube voltage of 120 kVp, automatic tube current, slice thickness of either 1.25 mm or 5 mm, and an interval of 0.625 mm. Because of variations in slice thickness and intervals across different images, uniform post-segmentation processing was applied. Thus, a slice thickness of 5 mm and an interval of 0.625 mm were obtained. Additionally, each patient underwent lumbar spine DEXA. Ground-truth labels for bone status were defined according to the WHO criteria using DEXA-derived T-scores: osteoporosis ($T\text{-score} \leq -2.5$), osteopenia ($-2.5 < T\text{-score} < -1.0$), and normal bone density ($T\text{-score} \geq -1.0$).

Image Segmentation

First, ROI segmentation was conducted using Insight Toolkit-Segmentation and Registration Toolkit-SNAP software. For each lumbar CT cross-sectional image, the trabecular ROI was delineated within the vertebral body, excluding the cortical shell, endplates, and posterior elements. Second, to minimize the time spent on manual annotation while ensuring accuracy, 30 patients were randomly chosen for segmentation by an orthopedic surgeon. Third, based on the ROI results from initial segmentations, the remaining images underwent semi-automatic segmentation. Two experienced orthopedic surgeons reviewed and modified the final ROI results. Finally, an orthopedic surgeon with 11 years of experience reviewed and

Table 1. Clinical Characteristics in the Training and Test Cohorts

Variable	Training cohort	Test cohort	<i>P</i> value
n	322	138	
Mean age (sd), years	58.17 (10.58)	57.51 (10.49)	0.502
Sex, n (%)			0.831
Female	211 (65.5%)	89 (64.5%)	
Male	111 (34.5%)	49 (35.5%)	
Mean weight (sd), kg	65.37 (11.83)	65.40 (10.94)	0.883
Mean height (sd), m	162.98 (7.87)	162.38 (7.93)	0.558
Mean BMI (sd), kg/m ²	24.52 (3.51)	24.73 (7.94)	0.575
WHO diagnosis, n (%)			0.495
Normal	118 (36.6%)	51 (37.0%)	
Osteopenia	133 (41.3%)	57 (41.3%)	
Osteoporosis	71 (22.0%)	30 (21.7%)	

m, meters; kg, kilograms; BMI: body mass index; n: the number of patients. All characteristics in Table 1 are summarized at the patient level.

revised the annotations to ensure accuracy of the final ROI outcomes.

Deep Learning Using DenseNet

We implemented the models in PyTorch. DenseNet-121 was selected as the backbone because its densely connected architecture promotes feature reuse and facilitates gradient propagation, which can support stable optimization and effective representation learning in medical imaging settings with limited-to-moderate sample sizes. Each vertebra was cropped to $128 \times 128 \times 64$ voxels according to the trabecular ROI mask, and voxel intensities were normalized to the range [0, 1] using min-max scaling. During training, data augmentation (random horizontal flipping and in-plane rotation) was applied to increase data variability and reduce overfitting. We trained two independent DenseNet-121 models with identical pre-processing and backbone architecture: one for continuous trabecular BMD regression and one for three-class bone-status classification (normal/osteopenia/osteoporosis). Both models were trained for 200 epochs with a batch size of 32, with shuffling enabled each epoch. Optimization used the Adam optimizer with an initial learning rate of 1×10^{-4} . Mean squared error (MSE) loss was used for regression, and cross-entropy loss with a softmax output layer was used for classification. Hyperparameters were prespecified based on commonly used DenseNet training settings for medical imaging and were kept fixed across all experiments. The held-out test cohort was used only for final evaluation and was not involved in model selection or tuning. All experiments were conducted on a workstation with an Intel Core i9-7900X CPU and an NVIDIA RTX 3090 GPU. Figure 2 illustrates the process of ROI image acquisition and the subsequent models development.

Model Validation

After supervised training, the trained model's performance was evaluated using a test dataset. No data from the test datasets were included in the training dataset.

External Validation

In this study, external validation refers to a clinically distinct cohort from the same institution (clinical/spectrum validation) rather than a multi-center external dataset. Within Group A of the external-validation cohort, participants were split at the patient level into training and test sets in an 8:2 ratio to prevent data leakage, so that all vertebrae from the same patient were assigned to a single set. We developed a fracture-risk prediction model using vertebra-level inputs with patient-level case-control labels. Only non-fractured L1–L4 vertebrae from fracture patients were included, and L1–L4 vertebrae from non-fracture controls were included. Fractured vertebrae were excluded to avoid fracture-related morphological changes in the trabecular ROI that could bias BMD estimation. Each included vertebra was labeled as 1 if it originated from a fracture patient and 0 if it originated from a non-fracture control.

Using Group A, we fitted a multivariable logistic regression model with this binary label as the outcome and age, sex, BMI, and DEXA-measured BMD as predictors. Model discrimination was evaluated by ROC analysis in the test set. In Group B, DEXA-measured BMD and deep learning-predicted trabecular BMD were entered separately into the same fitted model while keeping age, sex, and BMI unchanged to generate vertebra-level predicted probabilities. ROC curves were constructed for each BMD input and compared using the DeLong test for correlated AUCs.

The predicted probability should be interpreted as a vertebra-level estimate of association with fracture case status rather than a direct prediction of whether that specific vertebra is fractured.

Results

Patients

The training cohort comprised 211 women and 111 men, with an average age of 58.17 years and an average body mass index (BMI) of 24.52 kg/m². The test cohort comprised 89 women and 49 men, with an average age of 57.51 years and an average BMI of 24.73 kg/m². No significant differences were observed in sex, age, or BMI between the two groups ($P > 0.05$). Specific demographic data are presented in Table 1.

External Validation Cohort

We developed a vertebra-level fracture discrimination analysis based on the external-validation cohort defined by the

Table 2. Clinical Characteristics in the External Validation Cohort

Variable	Group A	Group A	Group B
	Training cohort	Test cohort	
Number of vertebrae	514	127	240
Mean age (sd), years	58.2 (10.8)	60.1 (9.3)	64.4 (8.5)
Sex, number of vertebrae(%)			
Female	327 (63.6)	76 (59.8)	170 (70.8)
Male	187 (36.4)	51 (40.2)	70 (29.2)
Mean weight (sd), kg	65.4 (11.2)	64.9 (10.3)	63.0 (9.5)
Mean height (sd), m	162.7 (7.9)	162.2 (7.6)	161.4 (7.5)
Mean BMI (sd), kg/m ²	24.6 (3.2)	24.6 (3.1)	24.2 (3.5)
Mean lumbar vertebral BMD (sd), g/cm ²	0.89 (0.16)	0.89 (0.15)	0.95 (0.15)
Predicted mean lumbar vertebral BMD (sd), g/cm ²	0.89 (0.16)	0.88 (0.13)	0.81 (0.16)

m, meters; kg, kilograms; BMI: body mass index; “Number of vertebrae” denotes the vertebra-level sample size included in the fracture-risk prediction analysis. “Sex, number of vertebrae (%)”, “Mean lumbar vertebral BMD” and “Predicted mean lumbar vertebral BMD” are summarized at the vertebra level. Unless otherwise specified, characteristics are summarized at the patient level.

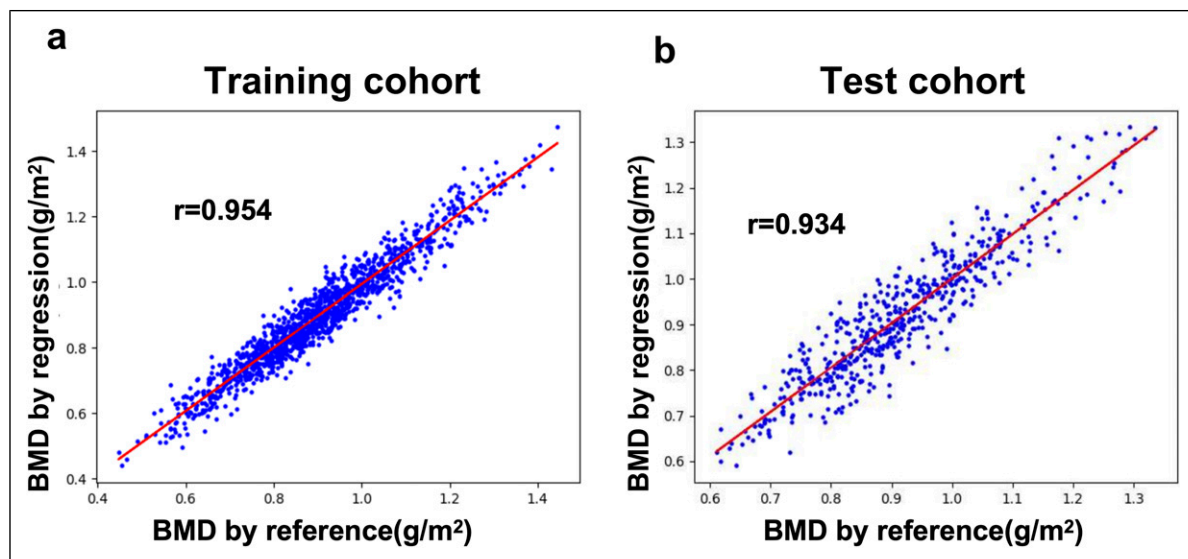
prespecified inclusion and exclusion criteria. Because CT-based measurements and deep learning–predicted trabecular BMD were extracted for each vertebra, analyses were performed at the vertebra level. Vertebrae from fracture patients were labeled as 1, whereas vertebrae from non-fracture controls were labeled as 0. Accordingly, Table 2 summarizes patient demographics, while the reported number of vertebrae indicates the vertebra-level sample size used for this analysis.

In Table 2, the external-validation cohort showed a higher mean lumbar DEXA-measured BMD than the training and test cohorts, likely because it included a substantial proportion of patients with lumbar osteophytosis and ligamentous calcification, which can artifactually elevate areal BMD on lumbar DEXA. By contrast, the lower mean deep learning–predicted trabecular BMD is consistent with the fracture-enriched case

mix of the external-validation cohort and the model’s focus on trabecular ROIs that are less susceptible to degenerative ossification–related bias.

Per Vertebra BMD Measurement Analysis

The training dataset had an average BMD of 0.898 g/cm², while the test dataset showed an average BMD of 0.919 g/cm². There was no significant difference in BMD values between the training and test datasets. Using DEXA-derived BMD as the benchmark, we assessed the BMD results from the regression model. The correlation between the lumbar BMD in the test set and the BMD predicted by the model was strong, with a coefficient of $r = 0.934$. The results are shown in Figure 3. In addition, we reported quantitative error metrics. In

**Figure 3.** Correlation plots of the average BMDs calculated by reference and automated regression in the training (A) and test (B) cohorts

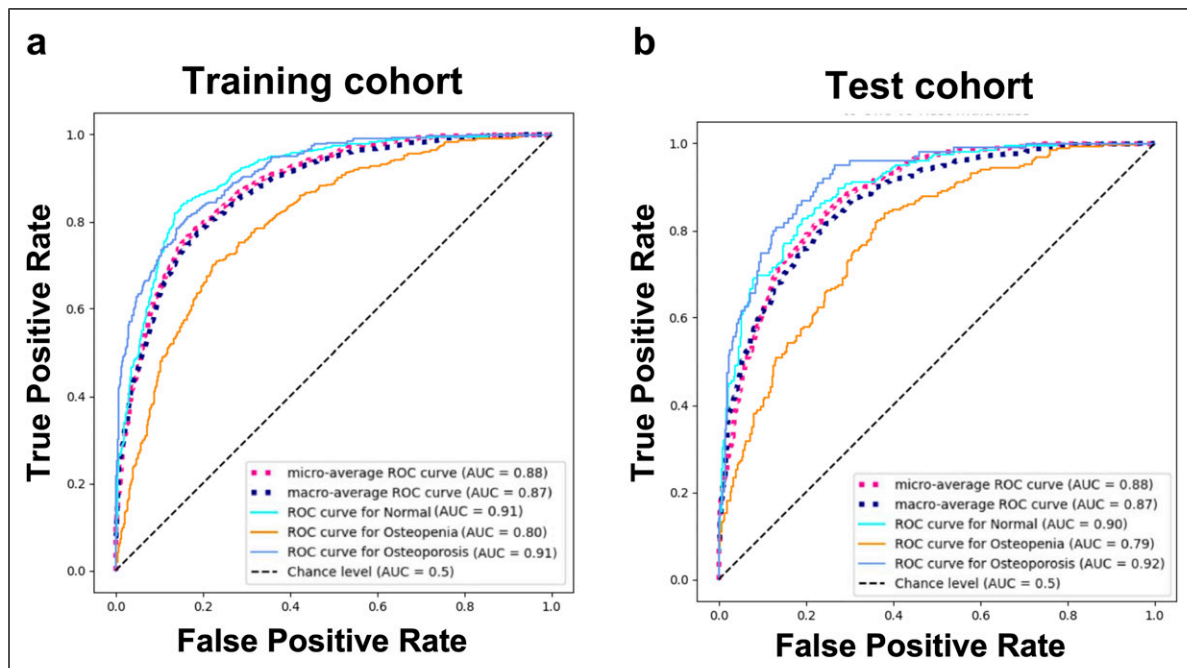


Figure 4. The ROC curves showing how the model predicts normal bone density, osteopenia, and osteoporosis in the training (A) and test (B) cohorts

the training cohort, the mean absolute error (MAE) was 0.0274 g/cm² and the root mean squared error (RMSE) was 0.0338 g/cm² (MSE = 0.0011). In the test cohort, the MAE was 0.0390 g/cm² and the RMSE was 0.0511 g/cm² (MSE = 0.0026).

Analyzing the Osteoporosis Classification Model

The training cohort consisted of 118 patients with healthy bone mass, 133 with osteopenia, and 71 with osteoporosis.

The test cohort comprised 51 patients with healthy bone mass, 57 with osteopenia, and 30 with osteoporosis. The predicted values demonstrated a strong alignment with the reference standards, as shown in the ROC curve in Figure 4.

In the test cohort, the AUC (area under the ROC curve) values for the normal bone mass and osteoporosis groups were notably high, at 0.90 and 0.92, respectively. In contrast, the AUC for the osteopenia group was lower, at 0.79. The results of the classification model are displayed in Figure 4.

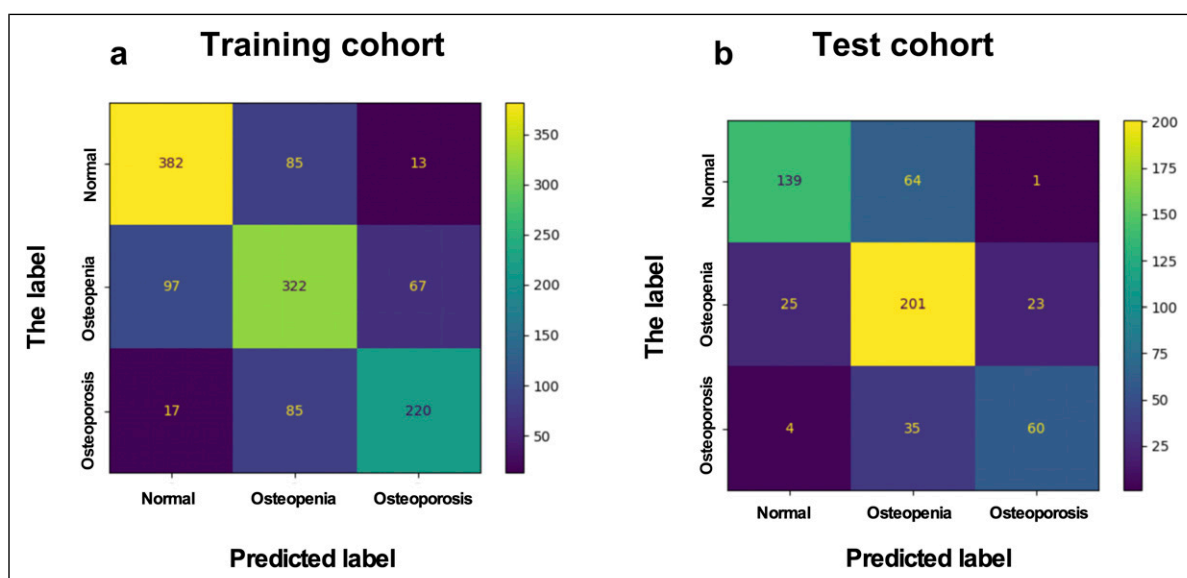


Figure 5. Confusion matrices of the three-class bone-status classification model in the training (A) and test (B) cohorts

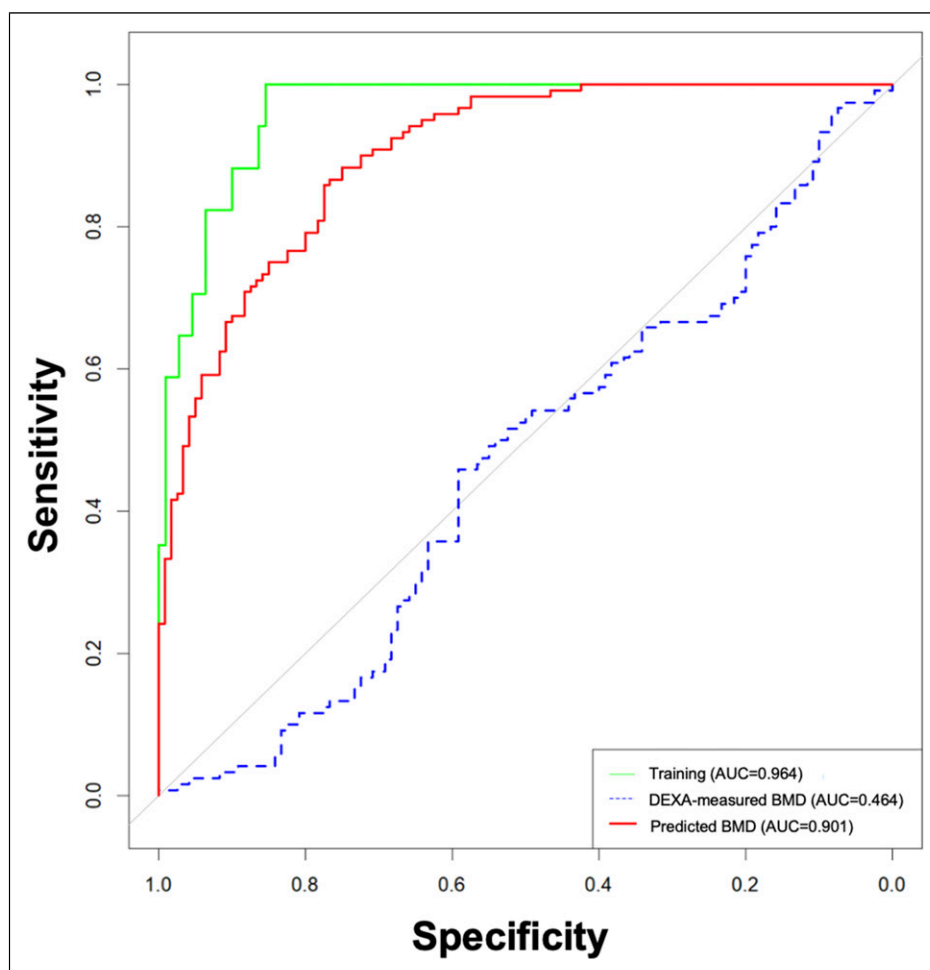


Figure 6. ROC curves comparing the discriminative performance of DEXA-measured BMD in non-hyperostotic patients, DEXA-measured BMD in hyperostotic patients, and deep-learning-predicted BMD in hyperostotic patients

Figure 5 presents the confusion matrix, which details the counts of true positives, false positives, true negatives, and false negatives for each vertebra's BMD.

Fracture-Risk Prediction Model

For the external-validation analysis, the ground-truth outcome was a binary case-control label assigned at the vertebra level, reflecting patient fracture case status: non-fractured L1–L4 vertebrae originating from fracture patients were labeled as 1, whereas L1–L4 vertebrae from non-fracture controls were labeled as 0 (fractured vertebrae were excluded). In subjects without hyperostosis (Group A), a DEXA-BMD-based logistic regression reference model showed high discrimination for this outcome ($AUC = 0.964$). In contrast, in the hyperostotic cohort (Group B), discrimination using DEXA-measured BMD declined substantially ($AUC = 0.464$). When deep learning-predicted trabecular BMD was entered into the same reference model, discrimination improved markedly ($AUC = 0.901$). The AUCs for deep learning-predicted trabecular BMD and DEXA-BMD were statistically

compared using the DeLong test, showing a significant difference ($P < 0.001$). The corresponding ROC curves are shown in Figure 6.

Discussion

In this study, a simplified method was developed to measure lumbar vertebral BMD and diagnose osteoporosis. A diagnostic accuracy study was conducted on adults who had undergone lumbar spine CT imaging, such as abdominal or lumbar spine CT, for other clinical indications. Despite some studies utilizing pelvic and thoracic CT scans to predict lumbar vertebral BMD,^{20,21} the lumbar vertebra remains an excellent site for observing bone loss across all skeletal structures. CT images can be used to measure the BMD of lumbar vertebrae and to identify patients with healthy bone mass, osteopenia, or osteoporosis. This method requires minimal training and time, making it suitable for prospective application across different departments, as well as for retrospective use by radiologists, orthopedic surgeons, and others. Additionally, because lumbar CT is routinely obtained before many orthopedic procedures,

our model may reduce the need for additional DEXA testing in selected patients by providing an opportunistic estimate of trabecular BMD and prompting targeted confirmatory evaluation when indicated. This approach incurs no extra costs, requires no additional time from the patient, involves no extra equipment, and avoids further radiation exposure before orthopedic surgery.

Jiang YW et al established a model by contouring the entire lumbar vertebrae to predict lumbar osteoporosis.¹⁵ Using lumbar CT images, we delineated ROIs within the vertebral trabecular compartment and developed the deep-learning models accordingly. It can be utilized to measure the BMD of lumbar vertebral and diagnose osteoporosis. Both the regression and classification models exhibited excellent predictive performance. Consequently, the regression model reflected the actual lumbar vertebral BMD more accurately, offering important clinical guidance for lumbar surgeries and BMD treatments. Xie et al established a model to differentiate between osteopenia and osteoporosis.²² A three-class classification scheme was implemented (normal/osteopenia/osteoporosis), providing a more granular stratification than prior binary models and thereby improving clinical interpretability. It allows physicians to perform additional DEXA scans to verify the presence of osteoporosis. In addition, the relatively lower performance observed for the osteopenia group may be attributable to its intermediate position between normal bone mass and osteoporosis, where cases near the diagnostic thresholds are inherently more prone to misclassification. These models may also contribute to lowering medical costs and minimizing radiation exposure.

This study has several strengths. First, because lumbar spine CT is frequently available from routine care, our framework supports opportunistic screening for osteoporosis and osteopenia without requiring dedicated imaging. Second, by using a strictly defined trabecular ROI with standardized preprocessing, our approach aims to improve measurement consistency and reduce susceptibility to variations in acquisition and reconstruction compared with relying on raw CT attenuation values alone. Third, we developed two complementary deep-learning models—one regression model that outputs a continuous trabecular BMD estimate and one three-class classification model (normal/osteopenia/osteoporosis)—thereby enabling both quantitative assessment and clinically interpretable stratification. Such stratification may facilitate clinically meaningful triage, prompting confirmatory osteoporosis evaluation and timely guideline-based prevention or treatment when indicated.

A key clinical rationale for our approach is the focus on trabecular-only CT regions of interest. In patients with spinal degeneration and hyperostosis, lumbar DEXA—typically acquired in the anteroposterior projection—can be confounded by osteophytes and ligamentous calcification, which may artifactually elevate areal BMD and T-scores and thereby weaken its association with fracture risk. Because our framework is based on trabecular-only ROIs, the network may

learn a comparatively “denoised” trabecular signal that is less affected by these degenerative findings. By training the model in a rigorously screened non-hyperostotic cohort where lumbar DEXA is less biased and by strictly confining the ROI to trabecular bone, the resulting CT-derived trabecular estimate may better approximate clinically relevant trabecular bone strength when applied to hyperostotic patients. We further conducted external clinical/spectrum validation in a clinically distinct cohort enriched for hyperostosis and ligamentous calcification, and within the same statistical reference framework, deep learning–predicted trabecular BMD showed substantially better fracture-case discrimination than DEXA-measured BMD in this subgroup, suggesting improved robustness to hyperostosis-related confounding.

Osteoporosis considerably affects the fixation implants used in spinal surgery, with severe osteoporosis potentially leading to postoperative screw loosening, non-union, and subsidence of interbody fusion devices.^{23,24} Therefore, spine surgeons should identify osteoporosis and tailor fixation strategies according to a patient’s BMD, and consider initiating anti-osteoporotic therapy before elective procedures when indicated. BMD can be evaluated using QCT, ultrasound, and DEXA, and the World Health Organization defines osteoporosis primarily based on DEXA measurements. Additionally, the use of QCT is increasing; it primarily targets trabecular bone to assess the bone quality.²⁵ However, its clinical use is frequently limited by high costs and the requirement for specialized training. The lumbar CT-based model offers an alternative method for assessing skeletal health.²⁶ This approach cannot replace DEXA; however, it can alert spinal surgeons to conduct further evaluations and potential treatment for osteoporosis. The proposed framework leverages routinely acquired CT scans from diverse clinical indications and incorporates a semi-automated approach for ROI delineation.²⁷ Therefore, it enhanced the consistency, accuracy, and ease of ROI delineation. In the future, we intend to develop one-click operations and other simpler methods for hospital use, offering a convenient and effective approach to assess lumbar vertebral BMD and screen for osteoporosis.²⁸ Furthermore, for delineating the ROI, we focused exclusively on the trabecular bone. To ensure that the DEXA results accurately reflected specific trabecular bone conditions, the patients were rigorously screened in both the training and testing cohorts, excluding the factors that could lead to inaccurate results. Consistent with QCT,²⁹ which primarily targets trabecular bone, the proposed approach estimates trabecular BMD from trabecular ROIs and may provide clinically meaningful information on bone quality. Additionally, it eliminated the complexities associated with QCT operations and processing, thereby reducing health costs.

This study has several limitations. First, it is a retrospective single-center study; larger prospective multi-center studies are needed to confirm generalizability. Second, our models were trained using DEXA-derived labels, and we did not include an independent quantitative reference standard such as QCT to

directly validate CT-derived trabecular BMD, particularly in patients with hyperostosis and ligamentous calcification; future studies should incorporate QCT-based validation and multi-center evaluation. Third, the deep learning models were developed using imaging-derived trabecular ROIs alone and did not incorporate demographic or clinical risk factors (eg, age, sex, BMI, comorbidities, medication use, or prior fracture history), which are important determinants of osteoporosis and fracture risk. Future studies should evaluate multimodal models integrating CT-derived metrics with key clinical variables, and consider statistical approaches that account for within-patient correlation when vertebra-level samples are analyzed. Despite these limitations, our results suggest that trabecular ROI-based deep learning from routine lumbar CT may provide a useful adjunct for osteoporosis screening and risk stratification, particularly in patients with hyperostosis where lumbar DEXA may be biased.

Conclusions

In conclusion, we developed and validated lumbar CT-based deep learning models for detecting lumbar vertebral BMD and screening for bone loss and osteoporosis in various scenarios, such as before lumbar surgery. These two models provide valuable information and facilitate surgical decision-making, enabling osteoporosis diagnosis, treatment, and prevention. Recommendations may include adequate dietary intake, fall prevention strategies, and pharmacological interventions, without incurring additional medical costs or radiation exposure.³⁰

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Ethical Considerations

This retrospective study was approved by the Ethics Committee and Institutional Review Board of The First Affiliated Hospital of Nanjing Medical University (Approval No. 2024-SR-132).

Consent to Participate

The requirement for informed consent was waived by the Institutional Review Board of The First Affiliated Hospital of Nanjing Medical University because the study utilized only anonymized retrospective data and imaging materials.

Author Contributions

J.F. and J.G. designed the study. J.W. collected data and wrote the paper. W.L. established the model. K.G. contributed to the writing. T.G., Q.W., T.Q., L.G., B.Z., and Z.Z. collected data. All authors reviewed the manuscript.

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Declaration of Conflicting Interests

The authors declared no potential conflicts of interest with respect to the research, authorship, and/or publication of this article.

Data Availability Statement

Due to privacy concerns, the data supporting the study's results are not publicly available but can be requested from the corresponding author upon reasonable request. The data are stored in a controlled-access repository at The First Affiliated Hospital of Nanjing Medical University.

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Appendix

List of Abbreviations

BMD	Bone mineral density
DEXA	Dual-energy X-ray absorptiometry
3D	Three-dimensional
2D	Two-dimensional
QCT	Quantitative computed tomography
ROI	Region of interest
MSE	Mean squared error
AUC	Area under the ROC curve
ROC	Receiver operating characteristic.